

Our responses to comments (in bold) are given as bullet points below.

Handling Editor

> Please follow the recommendations from reviewer 1 and Dr Sherratt about treating the two male morphs as distinct categories at the same hierarchical level as females. Also pay particular attention to placing your study in the broader evolutionary context, not only with respect to allometry, but also the evolution of sexual dimorphism.

- We have fully restructured the analysis so that females, brachylabial males, and macrolabial males are treated as three phenotypic groups within a single unified model. The revised model relates log forceps length to log pronotum length and includes all interactions among phenotypic group, diet, and density, as well as family-level random intercepts and slopes and a rearing-dish random intercept. We have removed the previous analysis that pooled the two male morphs and no longer estimate or interpret an overall “male” allometric slope. All comparisons among females, brachylabial males, and macrolabial males are now derived from the same model. We have also substantially revised the Introduction and Discussion to place the results within the broader literature on the evolution of sexual dimorphism, the exaggeration of male traits relative to homologous female traits, and the evolutionary relationship between females and alternative male phenotypes. The revised interpretation emphasizes the similarity between females and brachylabial males and identifies macrolabial males as the principal source of divergence in forceps allometry.

Associate Editor: Dr. Emma Sherratt

> There is evidently no homogenous allometric pattern within males, which means the sex-specific analyses are statistically misleading. As indicated by reviewer 1, it is more statistically sound to evaluate models with three groups: females and the two male morphs. As they say, “You can still ask whether males are different from females, but in a more nuanced way (e.g. are females different from macrolabial males or brachylabial males).”

- We agree and have replaced the separate sex-level and male-morph analyses with one unified model in which females, brachylabial males, and macrolabial males are treated as three levels of a single phenotypic-group factor. This model permits direct comparisons between females and each male morph, as well as between the two male morphs, while using a common residual variance and accounting for uncertainty within the same posterior distribution. The pooled male slope has been removed from the manuscript. The revised Results, figures, tables, and Discussion now focus on comparisons among the three phenotypic groups.

> On lines 178–182, this justification for the two separate analyses is not strong; averaging the males to a single slope tells us nothing about what is really going on—it is far more interesting to consider how the morphs evolve and differentiate via allometric scaling.

- We agree. The cited justification and the associated two-stage analytical approach have been removed. The revised Methods instead explain why the three phenotypic groups are analysed simultaneously. We no longer average across the male morphs or interpret a single male slope. This change reveals that females and brachylabial males generally have similar, strongly hypoallometric relationships, whereas the clearest departures from this shared pattern occur among macrolabial males.

> This is the change I expect in the revision.

- This change has been implemented throughout the manuscript and supplementary material. All condition-specific slopes and all contrasts among phenotypic groups are now estimated from the unified three-group model.

> Furthermore, the discussion is focused on static allometry. Yet one can make links to and thus broaden the scope by providing a broader statement about the origin and evolution of sexual dimorphism by considering the adage of males being exaggerated females—especially where one male morph resembles females. There is more literature on this than has been included in the discussion; I request improved scholarship in the revised discussion to incorporate this topic, which will increase the significance of this study from an evolutionary context.

- We have substantially broadened the Introduction and Discussion to connect the results to the origin and evolution of sexual dimorphism. In particular, we now discuss how exaggerated male structures evolve from homologous traits shared with females and how alternative male phenotypes can differ in the extent to which they diverge from the female condition. We use the similarity between females and brachylabic males, and the greater divergence of macrolabic males, to consider whether sexual dimorphism in this system is driven primarily by the differentiation of the exaggerated male phenotype rather than by a general difference between the sexes. We have added relevant literature on the evolution of sex-specific morphology, exaggerated male traits, alternative male phenotypes, and the developmental and evolutionary origins of sexual dimorphism. We have also softened the interpretation to distinguish the observed phenotypic pattern from conclusions about the historical evolutionary processes that produced it.

Reviewer 1

> I have read the revision of the manuscript. The authors have addressed almost all of my concerns, bar one. They continue to analyze males and females separately, when they know that the two male forms are significantly different in their scaling parameters. This generates two problems: First, you are comparing a biologically real female scaling against a chimeric quantity that conflates static allometry (within male morph) with the discrete difference between morphs. This means that a significant male/female slope difference may be an artifact of nuances of the number of males of each morph that you measured rather than any real difference in how the trait scales with body size. Second, and more fundamentally, the comparison is ill-posed: if males are genuinely dimorphic in scaling, there isn't one male slope to compare. The question “is the male slope different from the female slope?” doesn't have a single answer: it depends which male morph you are comparing females to. Further, there is also a reproducibility issue: because the pooled male slope depends on the proportion of macro to brachy males in the sample, and that proportion is likely an artifact of rearing or collection, the “male slope” isn't even a stable property of males. If you repeat the experiment and get a different proportion of each male morph, you will get a different slope.

- We agree with this criticism. We have removed the pooled female-versus-male analysis and no longer estimate an overall male slope. Females, brachylabic males, and macrolabic males are now analysed as three phenotypic groups in a single model. The revised primary model is

`log forceps length ~ log pronotum length × phenotypic group × diet × density + (1 + log pronotum length | family) + (1 | rearing dish)`.

Consequently, the estimated slope for each group represents within-group static allometry and is not influenced by the relative abundance of the two male morphs. The revised analysis directly compares females with brachylabial males, females with macrolabial males, and the two male morphs with each other within each developmental environment. We have revised the text throughout to eliminate references to a homogeneous male scaling relationship.

> As I indicated in my initial review, the authors should analyze female, macrolabial and brachylabial males as three groups. You can still ask whether males are different from females, but in a more nuanced way (e.g. are females different from macrolabial males or brachylabial males).

- We have implemented this recommendation. The revised analysis treats females, brachylabial males, and macrolabial males as equivalent levels of a phenotypic-group factor. Posterior contrasts are now used to evaluate differences between each pair of groups. These comparisons indicate that females and brachylabial males generally have similar allometric slopes, whereas the clearest differences are observed in macrolabial males. The Results and Discussion have been rewritten accordingly.

> On a related note, if the male morphs were separated only in the second analysis, where do the slope estimates for Figure 4 come from? Are they from two separate analyses: females slope estimated from the first female v. male analysis (model 1) and brachylabial and macrolabial slope estimated from the brachylabial v. macrolabial analysis (model 2)? They should be explicit about this in the figure caption: Parameter comparisons based on CIs across models can be unreliable, since the residual variance will be different in the two models.

- We agree that the previous presentation did not make the origin and comparability of these estimates sufficiently clear. This issue no longer arises in the revised manuscript because every slope shown in the revised figure—including those for females, brachylabial males, and macrolabial males—is estimated from the same unified model. The figure caption now explicitly states that all slopes and credible intervals are derived from one model. Formal conclusions about differences among groups are based on posterior contrasts from that model rather than on visual comparisons of credible intervals.

> The authors claim that there is a hierarchical structure in scaling: sex-specific differences in slopes combined with morph-specific differences in scaling sensitivity. As argued above, I am not confident that they are seeing sex-specific differences in scaling, since it is possible (but untested) that brachylabial males are not different from females. This reduces the claim that there is hierarchical structuring of weapon allometry. The clearest result therefore is that macrolabial males have different scaling than females and brachylabial males.

- We agree that our previous interpretation overstated the evidence for a general sex-specific difference. The revised three-group analysis directly tests the previously untested comparison between females and brachylabial males and provides little evidence for consistent differences between these groups. We have therefore removed the claim that the pattern consists of a general sex difference followed by differentiation between male morphs. The revised manuscript instead emphasizes that forceps allometry is characterized primarily by similarity between females and brachylabial males, with the most pronounced differentiation occurring in macrolabial males. References to hierarchical structuring have been revised to

avoid implying a homogeneous male slope or a universal contrast between females and males. The Discussion now identifies divergence of the exaggerated macrolabial phenotype, rather than sex alone, as the clearest biological result.